

Bayes classifier – Solution

- Step 1: Posterior Probability Calculation

- $\eta(s, h) = P(Y = 1 | S = s, H = h)$
 $= P(S + H + N > 5 | S = s, H = h)$
 $= P(N > 5 - (s + h))$, Since N is independent of S and H .

- Step 2: PDF of $X \sim \text{Exp}(\lambda)$ is given by

$$p(x; \lambda) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Since $N \sim \text{Exp}(\lambda = 1)$,

$$p(n) = e^{-n}, \quad \forall n \geq 0$$

The upper tail of the CDF $P(N > x)$

$$P(N > x) = e^{-x}$$

What if $x < 0$?

$$P(N > x) = 1$$

Bayes classifier – Solution

- Thus,

$$\eta(s, h) = P(N > 5 - (s + h)) = \begin{cases} e^{-(5-(s+h))}, & s + h < 5 \\ 1, & \text{otherwise} \end{cases}$$

- Step 3: Decision Boundary ($\eta = \frac{1}{2}$):

$$\begin{aligned} e^{-(5-(s+h))} &= \frac{1}{2} \\ -5 + s + h &= \ln(1) - \ln(2) \\ s + h &= 5 - \ln(2) \approx 4.31 \end{aligned}$$

- Thus, the optimal decision boundary is a line defined by:
 $s + h = 5 - \ln(2)$

Bayes classifier – Solution

- And the Bayes classifier is defined as:
- $\psi^*(s, h) = \begin{cases} 1 \text{ (pass),} & \text{if } s + h > 5 - \ln(2), \\ 0 \text{ (fail),} & \text{otherwise.} \end{cases}$
- If the student spends at least 4.31 hours per day studying the lecture and doing the homework, we can predict that the student will pass.

